LIVEPEER MARKET LANDSCAPE [*DRAFT*]

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1. Executive Summary

Purpose of This Report

Since its launch in 2017, Livepeer's mission has been to offer the world's open video infrastructure, building a respected and trusted brand rooted in decentralized video technology enabled by cryptoeconomic primitives.

More recently Livepeer has begun expanding its brand and technology into AI through various ecosystem projects and initiatives. However there has been a lack of a dedicated Go-To-Market effort to coordinate these efforts around a clear value proposition for the Livepeer Network. This report is intended to propose a scope and framework for such a Livepeer Network GTM effort by:

1. **Defining the Total Addressable Market (TAM) for Livepeer,** identifying:
    a. The global scale and impact of video
    b. The key problems and opportunities across video + AI ecosystems.
    c. Livepeer's unique capabilities within the ecosystem.
    d. Competitors and collaborators with similar offerings
    e. Ideal Customer Profiles for the Livepeer Network
2. **Conducting an Ecosystem Mapping Exercise** to segment potential customers and partners, recommending areas where Livepeer should:
    a. Maintain a baseline presence for credibility.
    b. Act as a bridge between AI and video communities.
    c. Lead and define new collaborative agendas.

Approach

***What is the potential reach of the Livepeer Network brand and technology*?**

Defining Livepeer's **TAM** (total addressable market) is important as it provides north stars to measure both the business success and social impact of the Livepeer Network, offering a conceptual frame of reference to inspire go-to-market activities including product development, business development, sales, marketing and community building.

This report provides a comprehensive analysis of where value sits in the global video economy. The starting point for the report is the premise that **end user video consumption is ultimately what determines demand for video infrastructure**: understanding consumer trends in how

video is both consumed and monetized is essential to develop successful strategies for scaled global usage of and sustainable growth in demand for the Livepeer Network.

Such an approach is even more critical given the opportunity for Livepeer to differentiate itself amidst an increasing global glut of generalized compute driven by the AI boom. Focusing on specialized compute for video workflows requires deep understanding of end use cases for video transformation and the future of video consumption.

There are 4 lenses for evaluating Livepeer's potential market :

TAM LENS	**Description**
Global Video Indicators	What is the global social and technological scale of video and how much of that activity takes place on the Livepeer Network?
Global Video Economy	How much total commercial value is created from the dynamics of supply and demand for video technology and how much of that value flows through the Livepeer Network?
Global Video Consumer Demand	How much commercial value is generated by the consumption of video and how much of that value is powered by the Livepeer Network?
Global Video Technology Supply	What is the cost of underlying technology required for video consumption?

Incorporating Video's AI Transformation

Throughout, the report highlights how AI is transforming the video economy, identifying key trends and patterns that create risks and opportunities for Liveeper.

1. Video is an exponential consumer of compute

Two converging forces are at play:

- First, online video consumption is at an all-time high and will continue to grow as more individuals around the world gain access to devices and new video experiences and formats come online.
- Second, AI is turning video into a GPU-intensive operation. Video is no longer just watched (where bandwidth is the key constraint) — it is being generated, enhanced, and

transformed in real time. AI video models (diffusion, autoregressive, world models) have 4–20x higher GPU cost than transcoding.

As a result, GPU hours dedicated to video are projected to triple from 35B in 2025 to 125B+ by 2030.

2. AI is not replacing conventional video but is overpowering it

AI is not replacing traditional, human-shot video — it is overwhelming it with scale, speed, and synthetic supply. Human video will remain uniquely valuable for emotional connection and cultural relevance, but by 2030 AI-assisted and AI-generated content will dominate total *volume* of creation:

- 10× more total video output, driven by automated editing, localization, and summarization
- 10–100× lower production cost due to AI-native workflows
- Infinite remixability across formats, languages, and styles
- Real-time personalization for every user, creator, or enterprise workflow

By 2030, synthetic or AI-augmented video could account for:

- 20–25% of all uploaded video
- 40–50% of corporate training & L&D content
- 30%+ of marketing and ad creative
- 10–20% of entertainment and short-form content

3. Video is becoming a multimodal compute object

For the first time, video creation is no longer limited to cameras or editing tools. AI systems can now generate or transform video from an ever expanding range of of media inputs :

- Text (descriptions, scripts, prompts)
- Images (reference frames, characters, environments)
- Speech / audio (narration, dialogue, emotional tone)
- Sensor data (motion, depth, environment capture)
- World-model states (agentic simulation → video output)

This turns video into the universal output layer for multimedia models — a far richer production pipeline than the traditional “record → edit → publish” video workflow.

4. AI creates a new economic layer around video data

The future of the video economy includes both high-value human video on one side, and abundant AI-generated video on the other. AI expands the global video economy with:

- Demand for high-quality video data grows for model training input
- Authenticity and watermarking becoming required primitives for trust and safety
- The potential for provenance to become monetizable, as all human-generated data becomes an increasingly scarce and valuable AI input

5. AI Accelerates today's concentration of power in the global video economy

As video workloads shift from GPU-assisted encoding → multimodal inference → generative video → agentic video production, the advantage sits with companies that own their own GPU fleets.

Today, over 80% of global video revenue flows through a small number of vertically integrated platforms that own both video audiences and video infrastructure. These hyperscale media platforms - including AWS, Google, Meta and Netflix have built monopolies on user attention and monetization. As a result, they have outsized budgets to spend on AI infrastructure and innovation that small to medium size media companies can't compete with.

Without a fullstack open alternative, the next era of video will be even more centralized than today.

2. Video's global scale and Livepeer's relative footprint

Video is the largest single driver of internet traffic, compute demand, and digital attention. The total number of global video users, scale of consumption, internet traffic and processing power all present the most ambitious lens for measuring Livepeer's growth and impact as the world's video open infrastructure:

- **How many people in the world consume video powered by the Livepeer network?**
- **How many minutes of video streamed globally are powered by the Livepeer network?**
- **How much video data passes through the Livepeer network?**
- **How much video is processed by the Livepeer network?**

Global Online Video Users

Year	Global Video Users (Estimated)	Notes
2015	~2.3B	Early mobile streaming era
2020	~3.8B	Pandemic acceleration
2025	~5.3B	Universal mobile video
2030 (proj.)	~6.0B	Near-global penetration

- In 2025, 91.8% of internet users worldwide watched digital videos weekly.
- By 2030, *almost every connected human* will watch online video daily.
- **This makes online video the largest media format in the world.**

Average Time Watching Online Video per Person per Day

Year	Minutes / Day	Notes
2015	~45 min	Early YouTube/social
2020	~70 min	Mobile-first + streaming
2025	~85 min	Short-form + live
2030 (proj.)	~120 min	AI-personalized and interactive video

Rising as social + AI video increases. Consumption is not saturating — short-form and AI-personalized video are increasing minutes watched, not cannibalizing other formats. Trillions of minutes streamed annually.

Video Share of Global Internet Bandwidth

Year	Total Video Data Transferred Globally (Absolute Traffic)	% of Total Internet Traffic	Notes
2015	~1.5 ZB	63%	Early HD era; Early broadband scaling
2020	~3 ZB	75%	Peak streaming adoption; 4K & mobile
2025	~6.5 ZB	82%	Short-form explosion; 4K

2030 (proj.)	~12 ZB	85%+	Personalized video and higher resolution; AI video, immersive, 8K

1 Zettabyte = 1,000 Exabytes; 1 Exabyte = 1 billion GB

AI video increases bandwidth demand because more content is being delivered, in more personalized and interactive formats.

Video Share of Global Compute

Year	**GPU Video Compute Hours (Billions)**	**% of Global GPU Compute**	**Notes**
2015	5 B	10%	Early streaming era; dominated by CPU-based transcoding with limited GPU use
2020	12 B	13.3%	Growth driven by 4K adoption, mobile video explosion, and platform-level standardization on GPU-assisted encoding
2025 (est.)	35 B	20.0%	Significant increase from AI video inference workloads (speech-to-text, moderation, embeddings). Generative video still niche but growing.
2030 (proj.)	125 B	20.8%	Video becomes one of the largest categories of global compute workloads. Growth driven by: user personalization, AI-generated scenes, upscaling, and multi-model inference.

The "**video compute multiplier**" is now the core driver of video infrastructure requirements : for every 1 minute of video consumed, there are 3–10 minutes of compute applied upstream. (Note: *video compute is d*oubling or tripling in absolute terms — but the total global compute market is growing even faster).

3. The Global Video Economy TAM

Video is one of the most valuable and rapidly expanding pillars of the digital economy, and the foundation of Livepeer's opportunity. This section presents a unified model of how economic value flows through the global video ecosystem today and how it expands over the next five years as AI fundamentally reshapes both video creation and consumption.

The **Total Addressable Market (TAM) for the Global Video Economy** captures the full commercial value generated by video worldwide. The model anchors on 2025 using aggregated, publicly sourced industry data, and projects to 2030 using a consistent CAGR framework that incorporates both stable structural drivers (advertising, subscriptions, commerce, SaaS adoption) and emerging AI-driven forces (synthetic video, multimodal interfaces, agentic media systems).

To reflect how the video ecosystem actually functions, the TAM is organized around **two complementary and interdependent components**:

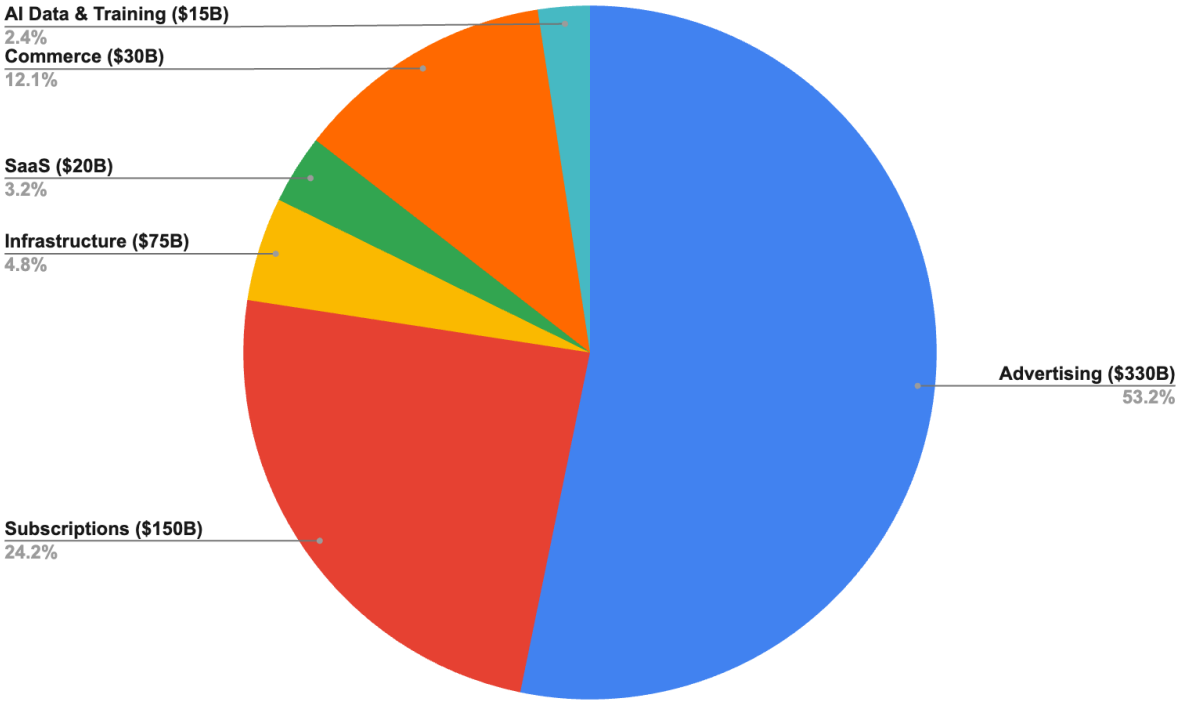
- **Demand Side** — all monetization generated inside online video platforms, including ads, subscriptions, commerce, and video SaaS revenue. This represents the cash flows created when consumers and enterprises engage with video.
- **Supply Side** — the infrastructure and computation required to produce, transform, and deliver video at scale. This includes both traditional operational video infrastructure (compute, storage, delivery, orchestration) and the rapidly growing upstream market for AI video data and model training. Together, these represent the cost base—and capacity—that enables the entire video economy to function.

These two sides form a single economic system: **demand expresses what users pay for; supply reflects what the ecosystem must compute to fulfill that demand.** Modeling them together provides a holistic view of the global video economy and a clear foundation for understanding how AI is rewriting its economics—shifting value toward compute-intensive pipelines, synthetic media generation, multimodal agents, and new forms of personalized, interactive video.

In total, the model estimates that the global video economy will reach **$620 billion USD in 2025**, expanding to **$900 billion USD by 2030**—driven increasingly by AI-native video workflows and the infrastructure required to support them.

Key Segments of the Global Video Economy

Video Economy TAM by Segment (2025, USD)



Segment	2025 (USD)	2030 (USD)	CAGR	Description
Advertising	$330B	$450B	6.4%	Ad-supported video across AVOD, FAST, hybrid ad tiers, live streams, and creator platforms—where value is driven by attention, impressions, targeting, and algorithmic yield optimization.
Subscriptions	$150B	$200B	6.0%	Recurring consumer payments for premium video access via SVOD platforms and hybrid subscription tiers that monetize through predictable monthly ARPU.
Commerce	$30B	$45B	8.3%	One-off consumer spend including TVOD, PPV events, rentals, gifting, tipping, and live commerce transactions—direct monetization tied to specific content moments or creator interactions.

SaaS	$20B	$35B	11.5%	Software, APIs, and tools used by developers, enterprises, and creators for production, editing, streaming, hosting, AI workflows, and video creation—capturing value through usage-based or subscription tools.
Infrastructure	$75B	$120B	9.8%	Operational infrastructure required to process, transform, and deliver video at scale across consumer, enterprise, and AI-enhanced applications. Includes production-time compute (transcoding, inference, generation, rendering), storage, delivery, and orchestration. Measures recurring OPEX consumed by platforms and services to run video workloads—independent of whether compute is rented or owned+operated in-house.
AI Video Data + Training	$15B	$50B	~27%	Upstream model-development spend for video as a core modality for multimodal AI systems: video dataset collection/labeling, preprocessing pipelines (frame extraction, deduplication, embeddings), and GPU-based training and fine-tuning of video, diffusion and agentic models. multimodal and generative video models.
TOTAL VIDEO ECONOMY TAM	**$620B**	**$900B**	~7.8%	

Key Macro Video Economy Insights

1. Infrastructure Bears the Load, Not the Value

- Infrastructure is only ~4.8% of the commercial value video economy, but determines 100% of video delivery
- Creates a fundamental asymmetry in the video economy: infra costs are small relative to revenue, but infra control determines who can enter, scale, or monetize video

2. The Video Economy is Highly Centralized

- A few giant platforms control both video audiences and infrastructure

    - More than 80% of global video revenue flows through a few vertically integrated platforms (YouTube, TikTok, Instagram, Netflix, Amazon, Twitch).
    - These companies own: the recommendation system; the encoding & compute stack; their own moderation & AI pipelines; private CDNs or custom delivery networks; the data layer
    - Creates end-to-end dependency where creators and developers can't switch vendors without losing distribution.
- Even when creators monetize, platforms capture the majority of economic value across ads, subscriptions, and commerce:
    - Subscriptions: Netflix, Disney, Amazon dominate.
    - Short-form: TikTok, Meta keep almost all ad & commerce revenue.
    - Creator payouts: YouTube ~45% platform take, others much higher.
    - Commerce video: TikTok/Amazon retain the majority of transaction value.
- Platforms' integrated infrastructure keeps their compute and delivery costs extremely low, allowing them to extract supernormal margins.
    - YouTube is the clearest example: delivery costs are low single-digit cents, while revenue per ad-view is orders of magnitude higher.
    - Compute costs are tiny relative to demand revenues across TikTok, Meta, Netflix, and Amazon.
- Infrastructure is itself centralized
    - Three hyperscalers (AWS, GCP, Azure) control ~70% of global cloud.
    - NVIDIA dominates >85% of AI accelerators, making GPU access a geopolitical choke point.
    - AI model training is concentrated in <10 global labs (OpenAI, Google DeepMind, Anthropic, Meta, X.ai, ByteDance, Tencent).
    - most video infrastructure runs on: AWS, GCP, Azure, Akamai, Cloudflare, Fastly
- This consolidation means:
    - Infrastructure margins shrink for everyone except hyperscalers
    - Open / decentralized infra struggles to scale into the market
    - Video APIs find it difficult to compete with vertically integrated incumbents whose marginal delivery costs → fractions of a cent
- Even creator-facing SaaS tools consolidate back into platform ecosystems. The more value these tools generate, the more platforms acquire and integrate them directly—further centralizing power.
    - Editing / VFX (Adobe, CapCut/ByteDance, Canva)
    - livestream tools (Streamlabs, Bending Spoons)
    - AI video & avatars (Runway, OpenAI, ElevenLabs)

3. Video infrastructure spend is growing faster than video revenue which *could* make centralized clouds less competitive

Year	Demand-Side Video Revenue (TAM)	Infrastructure Spend (Supply TAM)	Infra as % of Revenue	Notes

2025	$530B	$75B	14.1%	Video infra still predominantly encoding, CDN, storage; AI inference uplift begins.
2030	$758B	$120B	15.8%	Growth driven by AI inference, personalization, multimodal processing, real-time pipelines.
Delta (25→30)	+43%	+60%	Infra grows faster	Infrastructure grows faster than revenue due to AI's heavier compute requirements.

- Infrastructure spend (OPEX) grows *faster* than revenue as AI makes video more compute-intensive:
    - Multimodal inference (vision+text+audio)
    - Real-time personalization + generative video
    - World model pre-processing + simulation pipelines
    - Edge inference for live and interactive streaming
- Hyperscalers face structural pricing disadvantages relative to decentralized networks because of the amortization costs of investing in R&D and hardware (CAPEX)
- GPU pricing varies widely depending on hardware class, utilization, and procurement model. Published hyperscaler pricing for high-end instances often exceeds $4–5 per GPU-hour, while marketplace-based alternatives for non-frontier GPUs frequently clear in the $2–3 per hour range.
    - 1x H100 instance on AWS: **$4.10–$5+/hr**
    - Colos / alt-cloud pricing: **$2.00–$2.80/hr**
    - Bare GPU economics: **$0.80–$1.40/hr**
- While decentralized and marketplace-based systems introduce coordination and verification overhead, they can improve utilization and access lower marginal supply for workloads that do not require premium hardware. The economic opportunity lies in utilization efficiency and routing flexibility, not in eliminating costs entirely.
- Infra inflation could make Decentralized GPU markets more competitive
    - In theory, when infra grows faster than revenue, the market shifts toward price elasticity, pushing businesses to seek alternative supply; decentralized GPU markets could become the competitive release valve.

4. Global technology centralization is triggering demand for local sovereignty

Regulatory & Political Push for Local AI/Cloud Sovereignty

- Nations increasingly see compute and cloud as strategic national utilities—not commercial infrastructure
- Governments are explicitly rejecting dependence on foreign hyperscalers

Emerging Trend: Sovereign Video & Compute Infrastructure

- Growing concern that:
  - Video platforms (e.g., TikTok, YouTube) shape domestic culture and political discourse.
  - Compute monopolies can restrict access, shape markets, and create geopolitical imbalance.
- Countries are now exploring:
  - Local GPU clouds
  - National video platforms
  - Data trusts and public AI datasets
    Decentralized compute markets as neutral alternatives to foreign hyperscalers

🇪🇺 European Union	• EU Data Sovereignty Directive, GAIA-X, Digital Markets Act push for: ○ Local cloud control ○ Federated data infrastructure ○ Limits on US big-tech dominance • France & Germany backing large sovereign GPU clusters (GENCI, JUPITER).
Canada	• Canada and Germany are establishing a Digital Alliance on AI
🇮🇳 India	• Dedicated National AI Compute Mission building local GPU supercenters. • Mandatory data localization for key sectors (finance, telecom, health).
🇨🇳 China	• Fully sovereign GPU ecosystem (Ascend, Biren) + strict data residency laws. • Domestic social/video platforms are required to use Chinese-owned infra.
🇧🇷 Latin America	• Brazil's AI Sovereignty Strategy 2024 mandates local model hosting. • Regional governments investing in sovereign cloud via state-owned telcos.
🇺🇸 United States	• National Security framing: AI and compute capacity treated as critical infrastructure. • Increasing restrictions on exporting high-end GPUs.

Livepeer is uniquely capable of meeting these sovereignty needs

- Livepeer can become the "sovereign media compute rail" for nations, platforms, and enterprises seeking independence from cloud monopolies with:

    - Decentralized, jurisdiction-agnostic GPU supply
    - Open, verifiable compute suitable for regulated sectors
    - Local-first deployments (regions/cities/countries) without hyperscaler dependency
    - Built-in provenance + transparency for AI video generation
    - Lower-cost GPU access enabling markets where cloud costs are prohibitive

5. Open primitives are needed to enable a new wave of video software and applications to compete with hyperscale media platforms.

Today hyperscale platforms can internalize the entire video stack — compute, storage, delivery, AI, recommendation, ads, subscriptions — and therefore operate at a cost structure the long tail simply cannot match. There is no integrated, open alternative capable of competing with YouTube, TikTok, Twitch, Netflix, or Amazon. Without a full-stack open alternative, the next era of AI powered video will be *even more* centralized than today.

"Open" competitors can't compete without new primitives.

SaaS platforms and video tools (Mux, Brightcove, Vimeo Enterprise, etc.) are useful for developers, but they do not fundamentally change the competitive landscape because they do *not* provide:

- open distribution
- open monetization
- open attention markets
- value capture outside the hyperscaler ad + subscription rails

Open infra alone is insufficient — the commercial layer must be open too.

Tools alone cannot shift power because value capture sits in **attention, algorithms, and monetization**, not in the infrastructure APIs themselves.

The existing "open stack" (ffmpeg, gstreamer, OBS, WebRTC, Livepeer, etc.) is powerful but fragmented. It lacks the core primitives required for a full video ecosystem:

- recommendation and personalization
- audience aggregation
- monetization and payouts
- decentralized GPU supply
- open distribution networks

To enable a new wave of competitive video applications, the ecosystem needs open infrastructure + open commercial primitives — not just tools.

Livepeer is uniquely capable of serving as the coordination layer for the open video infrastructure stack and community.

4. Video Economy Demand Side TAM

Understanding Video Attention & Monetization

The demand side of the video economy measures the total commercial value of (revenue generated by) video creation and consumption. It includes what consumers, advertisers, and merchants pay for watching video, interacting with video, or conducting video-driven commerce. It also includes creator/developer SaaS tools which monetize video creation and streaming workflows in response to market demand for video content.

The model developed for this report estimates the Video Demand Side TAM to be **over half a trillion USD in 2025** at $530B, expanding to $730B by 2030. In this section we breakdown the 4 demand side segments of the video economy

Demand Segments

Segment	2025 (USD)	2030 (USD)	CAGR	Description
AVOD (all in-stream video ads)	$260B	$346B	~5.9%	Pure ad-driven monetization; revenue comes from impressions and programmatic demand. Ex: YouTube Ads, Facebook Watch, Pluto TV
FAST (free ad-supported TV)	$25B	$43B	~11.5%	Linear streaming monetized entirely by ads; resembles subscription TV but with ad-funded economics. Ex: Tubi, Pluto TV, Roku Channel
Hybrid (SVOD + Ads tiers)	$35B	$43B	~4.2%	Dual monetization blends recurring subscription revenue with ad inventory yield. Ex: Hulu, Disney+ w/ Ads, Netflix Basic w/ Ads
SVOD (pure subscription video)	$150B	$193B	~5.2%	100% recurring subscription revenue; no ads. Ex: Netflix (ad-free), Apple TV+, Amazon Prime Video
Live Streaming (creator/gaming)	$20B	$29B	~7.7%	Tool-driven and ad-supported ecosystem; creators monetized via subs, donations, ads; platforms take fees. Ex: Twitch, YouTube Live, Kick

Interactive Streaming (gifting, live commerce)	$15B	$34B	~17.8%	Transaction-driven model: gifting, tipping, live shopping; platforms take a cut of commerce flows. Ex: TikTok Live Shopping, Twitch Bits, YouTube Shopping
TVOD / PPV / Rentals	$5B	$8B	~9.9%	One-off transactional purchases: rent, buy, pay-per-view; revenue shared with rights owners. Ex: iTunes Rentals, Amazon Prime Video PPV, UFC PPV
Creator + Developer SaaS	$20B	$34B	~11.2%	Software + API layer that enables video production, processing, hosting, distribution, and creator workflows. Ex: Mux, Livepeer Studio, Cloudflare Stream, Agora, Streamyard, Vimeo Enterprise, Descript ; AI-first tools (Runway, Pika, CapCut).
TOTAL VIDEO DEMAND-SIDE TAM	$530B	$730B	~6.6%	

Who Pays Into the Demand-Side Video Economy

Consumers

- Subscriptions (SVOD + hybrid tiers), PPV/rentals (TVOD), tips, gifts/coins, channel memberships, premium features, live commerce purchases.

Advertisers & Brands

- In-stream video ads (AVOD, FAST, Hybrid), sponsorships, creator integrations, brand-funded content, and performance commerce ads.

Merchants

- Transaction fees and commissions in live commerce ecosystems, affiliate payouts, storefront integrations, and product placements.

Developers, Enterprises & Creators (SaaS / Tools)

- Payments for video software and tooling: encoding/transcoding, streaming APIs, conferencing, editing tools, production software, AI workflows, and enterprise platforms.

Where the Money Flows

Video Platforms & Content Distributors (Content Holders): ~55–65%

- YouTube, TikTok, Twitch, Netflix, Amazon Prime Video, Meta, Roku, FAST aggregators — platforms that host, distribute, or deliver the content (even if they don't own the underlying rights).

Publishers, Studios & Media Companies (Rights Holders) ~20–25%

- Entities that own the legal rights to the content — studios, production companies, sports leagues, labels, animation licensors. They earn licensing fees, revenue shares, royalties, and minimum guarantees.

Creators (Often Both Rights Holders and Content Holders) ~8–12%

- Individual creators who own the IP they produce and also act as distributors on platforms. They earn from ad share, tips, subscriptions, commerce splits, and affiliate revenue.

Commerce & Marketplace Platforms ~2–4%

- TikTok Shop, Amazon Live, Taobao Live, YouTube Shopping — the platforms facilitating transactions, taking a percentage of gifting, tipping, or commerce flows.

Video SaaS Providers ~5–8%

- Mux, Livepeer Studio, Cloudflare Stream, Agora, Vimeo Enterprise, Streamyard, Descript — tools and infra vendors that monetize developers, enterprises, and creators through software, APIs, and AI workflows.

The Hidden Value of User Data

There's another massive, largely invisible value layer: **user data embedded in advertising economics**.

Rather than representing a separate market, user data constitutes an implicit value pool embedded within existing advertising revenue. Identity, behavioral, and contextual signals improve ad targeting efficiency, attribution accuracy, and pricing power.

Analysis suggests that approximately **$100–150 billion of current global video advertising revenue** can be attributed to the yield uplift provided by data-driven targeting and measurement. This figure represents value at stake within existing monetization models, not additive market size.

Today, this entire value pool is captured by closed platforms—YouTube, TikTok, Meta, Amazon, Roku—with minimal economic participation for creators, viewers, or developers. AI-driven personalization amplifies this concentration, making user data the ultimate moat.

This creates a major opening for open data and open monetization primitives. If video applications, creators, or users can own and control the data that powers targeting, recommendation, and yield optimization, they can unlock a share of the “hidden” $150B value pool currently trapped inside walled gardens.

~~**Open video ecosystems that pair decentralized compute with user-owned data primitives have the potential to shift value toward creators, developers, and communities by making data a shared asset rather than an extraction mechanism.**~~

Consumer Verticals: Where Attention Goes

Looking at the demand side of the video economy also reveals what formats dominate video consumption. This section seeks to clarify what people watch, how large each consumer vertical is, and how these verticals map to what consumers and advertisers pay into the video ecosystem and where that money goes (the Streaming TAM) Each consumer vertical drives different:

- **viewing patterns**
- **bitrate / resolution requirements**
- **compute needs** (transcode, analysis, AI assist, live/interactive)
- **monetization models**
- **traffic patterns (live vs on-demand, bursty vs steady)**

Infra providers (Livepeer) scale with *volume* × *intensity*, and different video verticals produce very different combinations.

Attention vs Monetization (2025)

Vertical	**Share of Minutes (Attention)**	**Share of Revenue (of $510B)**	**2025 TAM ($B)**	**Notes**
1. Long-Form Entertainment (Movies/TV)	20–25%	34–36%	$173–184B	SVOD + hybrid; the largest monetization bucket; subscription-heavy.
2. Short-Form Social Video	35–40%	27–29%	$138–148B	Dominates minutes; monetization lags but rising via AI-driven ads.
3. Creator Live Streaming (non-interactive)	7–8%	5–6%	$25–31B	Driven by gaming + creator streaming; moderate ARPU.
4. Interactive Streaming (Commerce + Gifting)	5–6%	6–7%	$31–36B	Extremely high revenue/minute; live commerce + gifting.
5. Sports & Premium Live Events	3–5%	4–5%	$20–26B	Small in minutes but high-value rights + premium advertisers.

6. News & Information	8–10%	4–5%	$20–26B	High churn; lower CPMs; heavy real-time analytics workload.
7. Learning & Education	5–7%	~3%	$13–16B	Stable, stickier minutes; AI tutoring + enterprise L&D.
8. Corporate & Enterprise Video	<2%	1–2%	$5–10B	Small inside consumer TAM; most enterprise spend is off-TAM.
9. Immersive / XR / Spatial Video	<1%	<1%	$1–3B	Very early; extremely compute-intensive; strategically important.

Key Take-Aways from Video Demand TAM

1. Short-form dominates attention but lags in monetization.

Platforms are rapidly introducing AI-enhanced ads, commerce, and recommendation models to close this gap.

2. Long-form entertainment over-indexes in revenue.

Subscription economics + premium rights ensure high ARPU despite lower minutes compared to short form, live and interactive. Example: Netflix current bid to purchase Warner Bros.

3. Live & interactive formats punch far above their weight.

Although small in minutes, they deliver *extreme* revenue density and *extreme* infrastructure intensity.

4. Video is the new layer for commerce.

Interactive streaming for commerce and gifting is the fastest growing segment of the video economy.

While not counted as part of traditional global video TAM, video-driven commerce, measured in gross merchandise value (GMV), vastly exceeds the entire global video economy.
GMV = the total monetary value of all goods sold through an e-commerce platform or marketplace over a specific period. GMV is used to assess the overall sales performance and growth of a business.

	Global Video Commerce	**Global Video Economy**

	(GMV)	
2025	$1.3T	$620B
2030	$2.1T	$900B

This shows the potential for video not just as a 1 dimensional form of entertainment or attention, but as an interactive medium.

Global Video Commerce GMV Overview

Segment	2023 GMV	2025 GMV (Est.)	2030 GMV (Proj.)	Notes / Geography
TikTok Shop GMV	~$16B	$55–70B	$150–220B	Fastest-growing video commerce engine globally; strong in SEA, UK, US ramping.
Taobao Live (Alibaba)	~$600B	$750–800B	$1.0–1.2T	Largest livestream commerce platform in the world; China-dominant; includes video-driven sales.
JD Live + Douyin (China)	~$200–250 B	$300–350B	$450–600B	China's broader livestream commerce ecosystem beyond Alibaba.
Pinduoduo + Temu video commerce	~$30–40B	$60–80B	$120–160B	Mix of video feeds, embedded video ads, creator commerce.
Amazon Live GMV	~$2–3B	$10–15B	$40–60B	Growing with influencer storefronts; US-focused.
Shopee Live (SEA & LatAm)	~$5–7B	$15–25B	$50–75B	SEA has fast-growing livestream commerce adoption.
YouTube Shopping / Affiliate Video GMV	~$3–5B	$15–25B	$40–90B	YouTube's affiliate video + creator storefronts expanding.
Instagram + Facebook Shops Video GMV	~$10–15B	$20–30B	$50–80B	Meta shifting toward embedded checkout + creator video.
Western Markets Livestream Commerce (non-Amazon)	~$5–7B	$10–12B	$30–50B	US & EU far behind Asia; rapid growth expected post-2025.

Global Creator Gifting GMV (Virtual goods)	~$15–20B	$25–35B	$50–70B	Includes TikTok gifts, YouTube SuperChats, Twitch Bits.
Sports / Entertainment PPV GMV (ticketed video events)	~$4–5B	$6–8B	$10–15B	Boxing, MMA, concerts, fan events; GMV depends on PPV pricing & demand.
TOTAL Video Commerce GMV	~$900B	~$1.3T	~$2.1T	

5. AI is reshaping video monetization across every content vertical.

Content personalization, automated localization, enhanced ad relevance, and video-enabled commerce are increasing the **value per minute of video**.

a) Lower cost of production & operations

- Generative AI reduces post-production costs by 30–80% depending on workflow.
- Creators generate 5–10× more content using AI-first tools
- AI massively expands personalized video output

b) AI raises the economic output of every minute of video consumed:

- Ad relevance → higher CPMs (especially in short-form + AVOD)
- Improved subscription retention via personalization
- Higher conversion in interactive commerce via on-video recommendations
- Enterprises shifting to synthetic training/presentations instead of linear video

c) Greater personalization means more platform value capture

- The more AI tailors experiences to the user, the more value consolidates toward whoever controls data + attention + distribution + GPU compute.
- SaaS platforms increasingly depend on GPU access for their AI video features
- Platforms with vertically integrated infra (OpenAI, TikTok/ByteDance, YouTube/Google) can underprice or out-innovate
- Independent SaaS tools without infra struggle with margins

6. Demand TAM is ultimately what determines infra workloads.

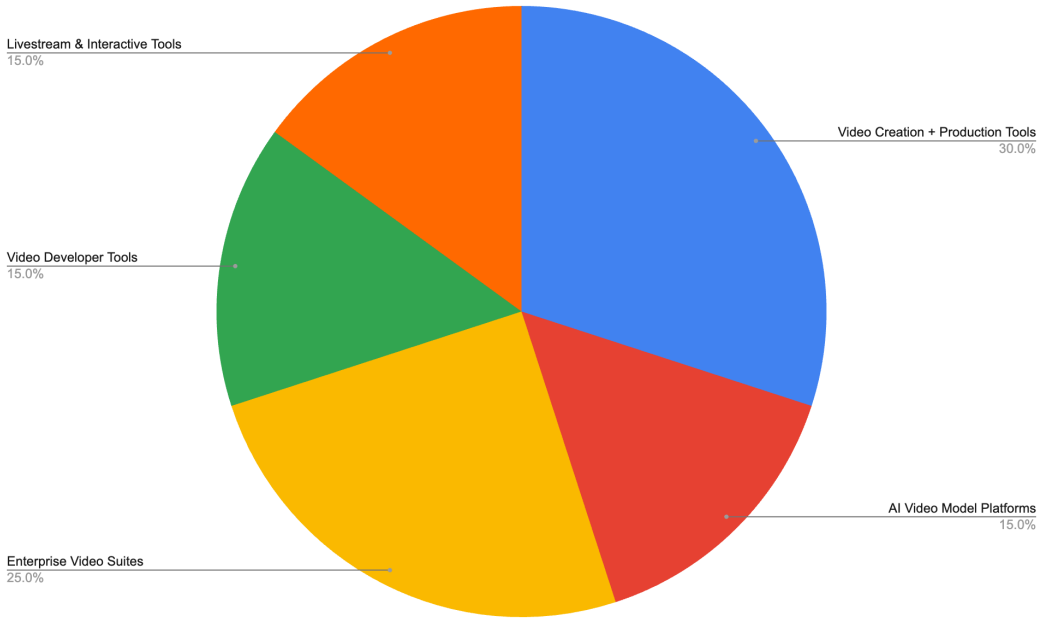
- The viewing mix sets compute patterns (AI analysis, generation), bitrate expectations, peak loads, and storage/delivery efficiency.
- This is why Hyperscalers are accelerating their dominance
- **Livepeer's open decentralized infra has the potential to serve SaaS as a neutral, scalable backend for the long tail / new wave of tools and applications.**

Video SaaS TAM

The Video SaaS TAM captures the revenue generated by **all software products that creators, businesses, studios and developers purchase to create, edit, manage, or integrate video.**

SaaS sits in the demand side layer where creators and developers generate the jobs that flow into that infrastructure. In other words, SaaS (eg Runway, Descript, Canva, Synthesia) aggregates consumer demand, while infrastructure (eg Livepeer, AWS Media Services, CoreWeave) fulfills it, making them economically distinct layers of the video ecosystem.

Video SaaS Segmentation ($20B TAM, 2025)



SaaS Segment	2025 TAM	2030 TAM	CAGR	Description
1. Video Creation & Production Tools (VOD)	$6.0B	$9.0B	~8.4%	Editing, clips, localization, captions; lightweight AI tools (Runway creator tier, CapCut, Descript).
2. Livestream & Interactive Tools	$3.0B	$5.0B	~10.8%	Multicam tools, overlays, multicasting, real-time engagement (Streamlabs, Riverside, Streamyard).
3. Enterprise Video Suites	$5.0B	$8.0B	~9.8%	Webinars, internal comms, training, avatar presenters, translation; high ACV (Vimeo enterprise, Kaltura, Synthesia enterprise).

4. Developer Video Platforms	$3.0B	$6.0B	~14.9%	APIs for processing, streaming, encoding, AI pipelines (Livepeer Studio, Mux, api.video, LiveKit, Cloudflare Stream).
5. AI Video Model Platforms	$3.0B	$7.0B	~18.5%	Access to models + builder environments (Runway, Pika, Krea, ElevenLabs for video, Fal, Replicate).
Total Video SaaS TAM	**$20B**	**$35B**	**11.5%**	

Video SaaS TAM Caveats

- Scope
  - Covers video-relevant AI model builder SaaS (fine-tuning platforms, hosted training/inference workspaces, model registries used for video + multimodal workflows), *not* the entire generic AI dev tooling market.
  - AI video analytics is a SaaS application layer that consumes video infrastructure and data; we include a conservative portion within our SaaS TAM but do not count the full vertical AI Video Analytics market as additive video economy TAM. The full vertical includes Large-scale **security / surveillance analytics. Smart city** video analytics, **Autonomous vehicle** perception tooling and **Industrial computer vision** platforms
- Double-counting
  - A significant share of Video AI Model platform revenue is passed through to infra/GPU providers. To avoid double counting, infra spend shows up only once in the Supply TAM;
  - The SaaS bucket is counted on the Demand side as the value-add service sold to end users.
- Data
  - Public revenue for emerging AI Video SaaS players in this segment is limited. The $3B → $7B path for AI Video model platforms is a top-down estimate, anchored to:
    - Overall (generic) AI dev tooling growth,
    - an assumed video/multimodal share of that market, and
      our model constraint that total Video SaaS = $20B → $35B.
- Category boundary:
  - Many vendors straddle SaaS and infra (e.g., Fal sells both SaaS-like model tooling and raw GPU time). In this model, we treat:
    - Their “platform / builder experience / APIs” revenue as SaaS
    - Their bare GPU resale margin as part of Infra (Supply TAM).

Key Video SaaS Trends

AI-native tools are taking share directly from traditional SaaS

- **AI Video Model Platforms are the fastest growing segment of the Video SaaS market,** as "Video SaaS" spend is migrating away from traditional creator tools:
    - Runway, Pika, and HeyGen have already displaced 20–40% of traditional video editing workloads among professional creators, based on publicly reported usage patterns.
    - Runway, Pika, and HeyGen grew 10× in usage in 2024–2025, despite flat growth in traditional creator SaaS categories (Adobe Express, Canva, Descript, Kapwing, VEED).
    - CapCut's (owned by ByteDance) global user base passed 350M MAU, driven largely by AI automation—not manual editing tools.
    - 40–60%+ of creators now rely on AI to auto-edit or generate content.
- **AI collapses front-end SaaS and developer workflows into models**

Opportunity for Livepeer:

- AI reduces reliance on frontend developer SaaS but dramatically increases reliance on compute, shifting value to GPU supply, low-latency inference, and scalable model routing.

5. Video Economy Supply Side TAM

The **supply side of the video economy** encompasses every upstream input required to produce, process, and operationalize video experiences. It's not content and it is not applications—it is the machinery behind them. Every stream, every upload, every inference call, every generative video output begins with supply-side computation: encoding, transformation, analysis, model development, and distribution. Supply-side spend grows reflexively with usage because it reflects *the physics of video itself*: more pixels, more modalities, more interactivity, and more intelligence all require more infrastructure.

Historically, the supply side was dominated by **traditional video infrastructure**—the compute, storage, bandwidth, and orchestration layers that power video workflows in real time. As video moved to the cloud, this became one of the most indispensable and least elastic categories of digital spend. Infrastructure is not discretionary; it is the substrate without which there would be no video economy.

Today, however, the supply side is expanding. Video has become a foundational input for multimodal AI systems, and this shift has created a new upstream category of supply-side demand: **AI video data and training**. Model developers now invest heavily in video datasets, preprocessing pipelines, and GPU-based video training workloads to build the next generation of generative, perceptual, and agentic video systems. These expenditures do not appear in

media revenue or operational compute—they are **production inputs**, analogous to encoding, storage, and delivery, but operating one layer earlier in the video value chain.

Together, **traditional video infrastructure** and **AI video data + training** form a unified supply-side ecosystem that exists to meet downstream demand for richer, smarter, more creative video experiences. One side serves video as it is used; the other shapes video as it is learned. Both are economically distinct from content, SaaS tools, and user applications, yet both are indispensable to the expansion of the video economy. As video shifts from a media format to an AI-native modality, the supply side becomes not just larger, but more strategically central—an integrated, compounding foundation for the entire computational video landscape.

	2025	**2030**	**CAGR**
Traditional Video Infrastructure TAM	$75B	$120B	~9.8%
AI Video Data + Training TAM	$15B	$50B	~22%
Video Supply Side TAM	**$90B**	**$170B**	**~13.5%**

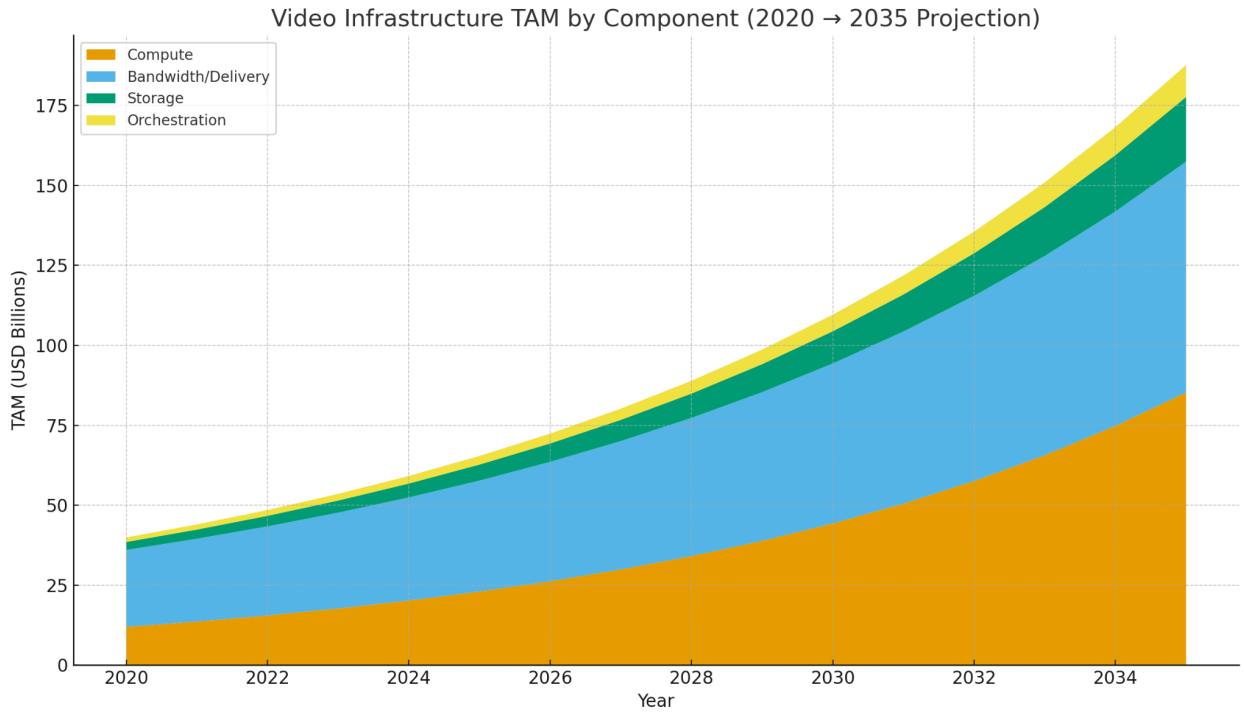
5.i. Traditional Video Infra Stack

Below is a detailed description of each infrastructure component that goes into creating and delivering a piece of video content. There are four primary components that together represent the entire supply-side cost base of the video ecosystem.

Video Infra Component	**Description**	**Includes**	**Cost Unit / What Drives Costs**
Compute	Workloads that actively transform, generate, or analyze data — tasks where CPU/GPU cycles perform meaningful processing.	• Transcoding / re-encoding video • AI inference or model evaluation • Video generation, rendering, enhancement • Encryption, watermarking, verification	$/minute processed or $/GPU-hour Costs scale with model complexity, resolution, and real-time requirements

Bandwidth / Delivery	Transport and real-time delivery of video from origin → viewer; routing, caching, and optimizing quality/latency.	• Network I/O & bandwidth delivery • Caching (read operations) • Edge optimization (latency, QoS) • Latency routing • Low-latency transport (WebRTC, QUIC, LL-HLS)	$/GB delivered Costs scale with audience size, bitrate, geographic distribution, and QoS requirements
Storage	Persisting, retrieving, and verifying video assets at rest for playback or AI workflows.	• Object storage (S3, GCP, Backblaze) • Cold archive (Glacier, Google Archive) • Origin storage for CDNs • Hot / warm caches for streaming and AI	$/GB/month Costs scale with library size, retention policy, and access frequency
Orchestration	Cross-cutting systems that coordinate workflows across compute, storage, and delivery.	• Logging & observability • Monitoring & QA • Access control / token gating • Metadata & identifiers (ex: CIDs) • API gateways / auth layers • Workflow engines & job scheduling	% System overhead Costs scale with workflow complexity, automation needs, and integration depth

Video Infra TAM (by Segment)



Component	2025	2030	CAGR
Compute	$23B	$45B	14%
Bandwidth / Delivery	$34.7B	$50B	7.6%
Storage	$5B	$10B	15%
Orchestration	$2.7B	$5B	14%
TOTAL VIDEO INFRA TAM	**$75B**	**$120B**	**9.8%**

Bandwidth / Delivery

Historical patterns

- Historically the largest component of video infra (≈60–70% of total spend).
- Highly optimized: unit cost declines ~8–12% annually due to CDN efficiency, peering, and hardware acceleration.

- Total bandwidth demand still rises as short-form, high-bitrate, and live video scale globally.

Emerging dynamics

- AI increases compute *far more quickly* than delivery; many AI transformations happen before video reaches the viewer.
- Bandwidth remains the largest absolute spend category, but its share of infra declines (70% → ~60% by 2030).
- Delivery becomes the “stable base layer” while compute becomes the growth engine.

Storage

Historical patterns

- Predictable, low-volatility cost base (~8–10% of infra)
- Long-term archiving + cheap cloud storage kept growth modest

Emerging dynamics

AI radically expands the role of storage:

- Embedding stores
- Intermediate ML assets
- Multi-variant outputs for personalization
- Synthetic training datasets
- World model corpora

Storage becomes a **strategic** part of the **AI data supply chain**, not just video archiving.

Orchestration

Historical patterns

- Small, stable category (~5% of infra).
- Grew slowly alongside feature complexity (DRM, analytics, control planes).

Emerging dynamics

Orchestration grows faster than bandwidth/storage because modern video pipelines are model-defined, policy-defined, and latency-defined.

AI transforms video workflow orchestration into a *high-growth* category because:

- AI pipelines introduce multiple upstream stages (understand → enhance → generate → verify → deliver).

- Model routing, retries, and orchestration become mandatory.
- Real-time inference (avatars, filters, overlays) increases application-layer logic per frame.
- Verification & provenance (C2PA, watermarking, hashing) become required steps in AI-generated or AI-augmented video workflows.

Growing Supply of Heterogeneous GPUs

The unprecedented level of speculative industry investment in AI hardware (GPUs) underscores the emerging market potential for hardware orchestration to ensure investors and businesses get the most of the data center building boom and hardware coming online. As Meltem Demirors, Founder of Crucible Capital writes: "hardware will need software to monitor, maintain, orchestrate, and optimize hardware utilization and capex monetization".

The current AI cycle is characterized not by uniform GPU oversupply, but by growing heterogeneity in compute availability. While frontier datacenter accelerators remain constrained and premium-priced, non-frontier and secondhand GPUs are increasingly liquid and price-discoverable through marketplaces and alternative providers.

This divergence matters because many video workloads do not require frontier hardware. As interchangeable GPU supply expands, differentiation shifts away from owning compute toward coordinating execution across heterogeneous resources.

To the extent that specialized video hardware orchestration can optimize both newer and older GPUs, this is attractive for investors monitoring GPU deployment and rate of depreciation https://x.com/Melt_Dem/status/1995624639528403180?s=20 See more on Crucible Capital's DePIN thesis here.

Compute

The Video Infra Architecture Shift: From Transport to GPU Execution

Historical Patterns

- Video infra economics has historically centered around: CDN (transport) + encoding (compute) + files (storage).
- In 2015, compute was only ~15% of total video infra spend, almost all of which was transcoding.
- Over the last decade, compute has been the fastest-growing infra category (17–20% CAGR).

Emerging Trends

- By 2030, compute will grow to reach ~30% of all video infra spend.
- The core video infra bottleneck shifts from throughput (bandwidth) to latency-constrained GPU scheduling (compute).

- Video infrastructure becomes model-driven software pipelines running continuously on GPUs, shifting from a transport problem to a GPU job execution problem.

Total Video Compute TAM: $23B (2025) → $45B (2030)

Video Compute Multiplier

- AI increases compute requirements by 3–6 times per minute of video. Every stream is more likely to require:
    - Captioning, detection, face/object tracking
    - Super-resolution, reframing, enhancement
    - Personalization (variants per viewer)
    - AI agents adding overlays or commentary in real-time
    - World-model semantic understanding & simulation
- Compute TAM grows 2–3× faster than all other infra components.
- The total cost of any given stream depends on its compute:density ratio

Video Compute Workloads

Legacy TAM models treated compute cost as “how many minutes need to be transcoded.” In the AI era, this is no longer accurate.

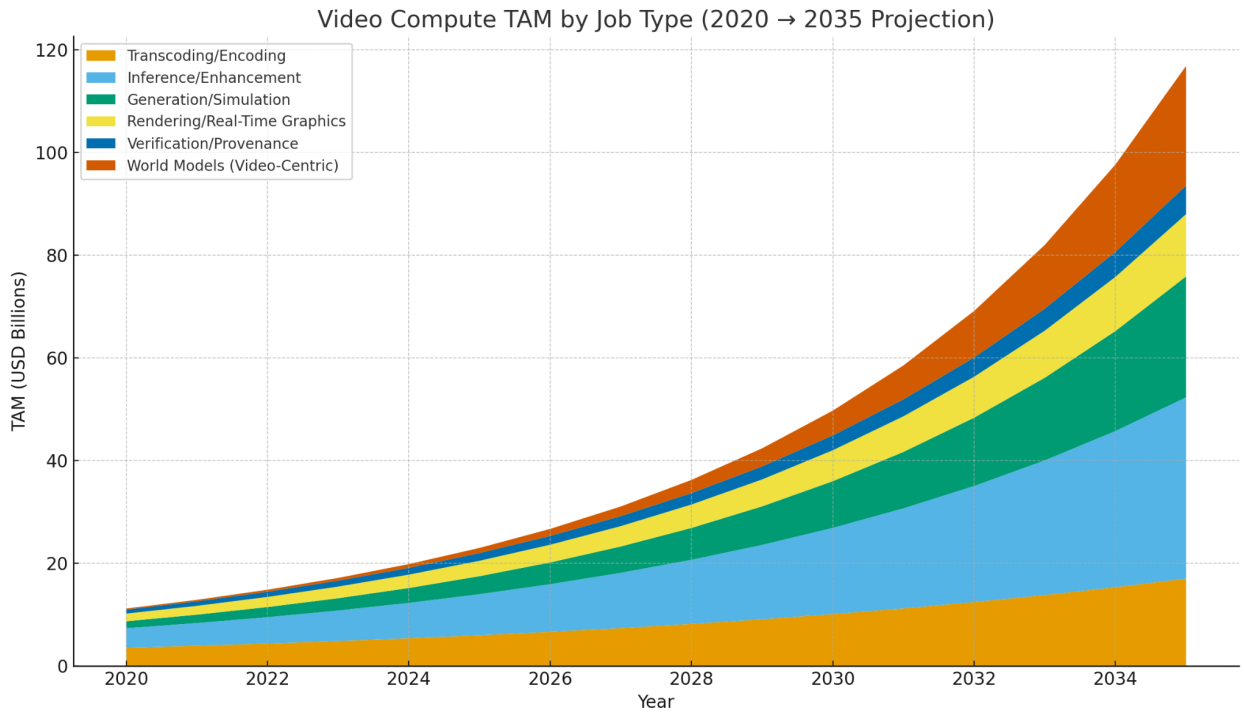
There are now **distinct classes of compute jobs**, each with different:

- hardware requirements
- latency constraints
- batching / parallelization patterns
- cost per minute
- scaling dynamics

Understanding these classes is essential for:

- sizing Livepeer’s opportunity
- designing product surfaces
- pricing workloads
- identifying partnership vectors

Video Compute TAM by Workload (Job Type)



Job Type	Description (What This Compute Workload Does)	2025 TAM (USD)	2030 TAM (USD)	CAGR
Transcoding / Encoding	Converts video into different codecs, resolutions, bitrates, and containers for delivery. Required for every stream. Mostly non-AI.	$6B	$10B	11%
Inference / Enhancement	AI models that understand or improve video: captioning, detection, moderation, embeddings, upscaling, color correction, reframing. Used in both VOD and live pipelines.	$8B	$17B	16%
Generation / Simulation	Creates new video pixels or frames from text, images, prompts, or partial observations (diffusion, VAE, autoregressive video). Includes NeRFs and 3D scene synthesis.	$3.5B	$9B	21%

Rendering / Real-Time Graphics	Frame-by-frame GPU rendering for interactive, latency-sensitive applications: avatars, filters, XR/AR, virtual production, cloud gaming.	$3B	$6B	15%
Verification / Provenance	Validates, audits, and traces video integrity: hashing, watermarking (C2PA), content authenticity, compute proofs, and analytics.	$1.5B	$3B	14%
World Models (Video-Centric)	Large multimodal models that learn physics, causality, and environment dynamics from video. Used for agents, robotics simulation, autonomous systems, and generative environments.	$1B	$5B	37%
TOTAL VIDEO COMPUTE TAM		**$23B**	**$45B**	**14%**

Video Compute TAM by Latency Requirement

Latency is the strongest determinant of cost, hardware requirements, and infrastructure architecture. However, **"Real-time" is not a job type. It is a latency requirement applied across video workloads.** Real-time can apply to *many different* functional workloads. It is a timing constraint (SLA requirement), not a functional category. For example:

- Real-time inference
- Real-time generation
- Real-time rendering
- Real-time world-model evaluation
- Real-time transcoding (live video)
- Real-time analytics (policy / safety)
- Real-time verification (watermark detection for livestreams)

Using our TAM model we can also estimate all workloads by latency requirement, summing portions of each functional category:

Video Compute TAM by Latency Tier

Latency Tier	Definition	2025 TAM ($B)	2030 TAM ($B)	CAGR

Real-Time (sub-250ms)	Must produce output instantly per user or per frame	8B	22B	22–25%
Near Real-Time (seconds)	Must process output quickly, but not instant	7B	13B	13–14%
Batch (minutes–hours)	No latency requirement; throughput optimized	8B	10B	4–5%
TOTAL VIDEO COMPUTE TAM		**$23B**	**$45B**	**14%**

Real-time compute (sub-250ms) is the **fastest-growing segment** of the entire video infrastructure economy, driven by:

- AI avatars
- interactive agents
- real-time world-model queries
- live augmentation of video
- real-time effects and rendering
- interactive livestream commerce
- multiplayer + simulation workloads

Deep Dive: Compute Jobs by Latency Requirement

Real-Time (<250ms)

Compute must operate under strict per-frame or per-interaction SLAs (often <50ms per frame or <100ms round-trip).

This is the **highest-cost, least flexible** compute class.

Characteristics

- Zero batching — every frame processed “hot.”
- Requires GPUs/ASICs on standby.
- Highest operational overhead: speed, scheduling, determinism.
- Needs edge or geographically distributed GPU nodes.
- Strict QoS, jitter control, and priority routing.

Examples

- Live inference (face/pose/gesture tracking)
- AI avatars, VTubing, lip-sync
- Real-time translation/captioning
- Live segmentation, background removal
- AR filters, VFX overlays
- Interactive agents reacting to live video
- Cloud gaming rendering

Hardware Profile

- High-end GPUs with strong tensor cores (4090, L40, A40/A100)
- Low-latency memory, optimized kernels

Near Real-Time (0.5–2s)

Fast enough to feel instantaneous, but not frame-by-frame.

Latency allows **small batching**, significantly lowering costs.

Characteristics

- Suitable for clip-level or segment-level processing
- Runs well in regional datacenters or cloud clusters
- Parallelizable pipelines with moderate time budgets

Examples

- Moderation (NSFW, copyright)
- Speech-to-text, captioning
- Upscaling, reframing, stabilization
- Watermark verification
- Just-in-time transcoding
- Ad personalization / dynamic reframing
- Multi-angle stitching for live events

Hardware Profile

- Mid–upper-tier GPUs: A10, L4, 3090/4090
- CDN or regional GPU clusters / datacenters acceptable (not edge-dependent)

Batch (Minutes–Hours)

No user-facing latency requirement.

Optimized for **throughput, cost efficiency, and massive parallelism**.

Characteristics

- Ideal for running when GPU capacity is cheapest
- Maximum batching and job parallelization
- Often uses spot/pre-emptible instances
- Heavy reliance on object-storage-to-compute pipelines

Examples

- Full encoding ladders (VOD transcoding)
- Bulk moderation, embeddings, classification
- Model training & fine-tuning
- Large-scale frame interpolation
- Offline generative video
- Dataset preprocessing & synthetic data generation (especially relevant for world models)

Hardware Profile

- A100/H100 clusters, TPUv4
- CPU-heavy preprocessing pipelines
- Cost-sensitive deployment strategies

Key Video Compute Insights

1. **Transcoding becomes a smaller % share of a much larger TAM, but remains a larger absolute market.**

   Every AI video workflow still ends in transcoding in order to steam at scale. Transcoding continues to grow steadily at 11% CAGR.

2. **Real-time compute is the fastest-growing and most constrained segment of the video infrastructure TAM**

   It requires sub-250ms latency, jitter control, and edge proximity — areas where centralized clouds are weakest and where Livepeer's globally distributed GPU network has a structural advantage.

3. **Near-real-time workloads form a large, cost-sensitive middle market.**

   Moderation, speech-to-text, clip-level inference, and enhancement benefit from Livepeer's ability to **route jobs across heterogeneous GPUs within the network** based on latency, hardware capability, and availability.

4. **Batch compute grows slowly in revenue but dominates total GPU hours**

   For offline generative video, interpolation, dataset processing, and VOD pipelines, Livepeer's low-cost, heterogeneous GPU pool can compete on price-performance where cloud clusters are overbuilt or overpriced.

5. **The rise of real-time AI makes GPU coordination and orchestration the new infrastructure bottleneck.**

   As video becomes model-driven and latency-defined, the ability to route workloads across diverse GPU nodes — matching jobs to the optimal node based on location, VRAM, and performance — can become a competitive moat.

Video Compute Opportunities for Livepeer

Livepeer's strongest opportunities concentrate in protocol-level routing for compute workloads that operate *on the video itself*:

A. Real-Time & Near-Real-Time Video Workflows

- Hyperscalers struggle with geography, jitter, and proximity
- Livepeer's distributed architecture is naturally optimized for low-latency GPU selection

B. Batch AI Video Generation

- Price-sensitive workloads (diffusion, frame interpolation, dataset prep) can run on surplus, heterogeneous GPUs

- Livepeer's network can undercut centralized clouds on cost and throughput for these workloads

C. *Transcoding as a stable economic floor for network supply*

- Real-time AI workloads are erratic and bursty, transcoding provides predictable, recurring demand that monetizes idle GPU capacity, keeps nodes online and ensures reliable global coverage

5.ii. AI Video Data + Training (Computational Video) TAM

AI Video Data + Training is treated as a distinct supply-side segment to capture the rapidly growing economic activity tied to video's role in model development, rather than in production or distribution. As AI systems increasingly learn from and generate video, a new category of upstream spend has emerged—one that is not reflected in traditional video TAMs but is essential to understanding the full computational footprint of the video economy.

The AI Video Data and Training TAM is projected to grow from **$12–18B in 2025** to **$40–60B by 2030**, driven by increasing demand for video datasets, video-specific preprocessing workflows, and training of large video and multimodal models.

Why AI Video Data + Training Is Included in the Video Economy TAM

Video is becoming a core training modality for AI.

Next-generation AI systems—multimodal LLMs, diffusion models, world models, robotics agents—depend heavily on video as a foundational learning signal. This has created substantial new upstream demand for:

- dataset collection and labeling
- video-specific preprocessing
- GPU-based model training and fine-tuning

Although these expenditures come from AI labs rather than media platforms, they originate from video workflows and represent real economic activity driven by video's growing importance in AI.

It is not captured in traditional video TAM definitions.

Classic video TAMs measure monetization (ads, subscriptions, commerce) and operational OPEX for serving video (compute, storage, delivery). They do not capture:

- model training
- dataset creation
- preprocessing pipelines
- training-specific GPU and networking demand

Yet these activities are directly tied to the rise of multimodal video models.

It is strategically adjacent to Livepeer's opportunity.

While Livepeer is not a training infrastructure provider, upstream AI video workloads:

- expand the total video compute opportunity
- increase the demand for preprocessing, inference, and dataset operations
- create new markets for verifiable compute, provenance, and rights-cleared datasets
- shift more AI and video value into distributed compute systems

Treating AI Video Data + Training as its own TAM segment gives a more accurate picture of how the Computational Video Economy is emerging at the intersection of video and AI.

To avoid double-counting, it is separated from the Video Infrastructure TAM.

Operational video infrastructure covers compute that acts directly on video streams (transcoding, inference, generation, rendering). AI Video Data + Training covers compute that

acts on models, not on user-facing video. These economics are fundamentally different and do not overlap.

Why AI Video Data + Training Is Its Own Category

It includes both OPEX and CAPEX, unlike Video Infrastructure TAM.

Model training requires:

- dedicated clusters
- H100/A100 GPUs
- high-bandwidth fabrics (NVLink, InfiniBand)
- parallel storage systems

This combination of capital expenditure + recurring compute OPEX is structurally distinct from the OPEX-only nature of video infrastructure workloads.

Training workloads differ fundamentally from operational video workloads.

Property	Training Workloads	Operational Video Workloads
Temporal pattern	Episodic, spiky	Steady, demand-driven
Synchronization	Highly synchronous	Loosely coupled
Hardware	Homogeneous H100/A100	Heterogeneous GPUs acceptable
Networking	NVLink, InfiniBand	Commodity internet routing
Runtime	Multi-day jobs	Short-lived, stateless jobs
Purpose	Model development	Production video compute

Livepeer is optimized for distributed, heterogeneous, stateless video compute, not HPC training clusters.

AI Video Data + Training TAM by Segment

Segment	Description	2025 TAM	2030 TAM	CAGR
A. Dataset Collection & Annotation	Human/automated video labeling: bounding boxes, segmentation, action tagging, QA. Examples: TELUS Digital, Scale AI, Appen, Sama, Labelbox.	$4B	$10B	~20%

B. Dataset Preprocessing (Video-Specific)	GPU/CPU parallel preprocessing: frame extraction, resizing, slicing, deduplication, pseudo-labeling, embeddings, formatting. Examples: Lambda, CoreWeave, RunPod, in-house AI labs.	$5B	$13B	~21%
C. Video Model Training & Fine-Tuning	Training + fine-tuning video models (diffusion, autoregressive, NeRF, multimodal transformers) on H100/A100 clusters. Examples: AWS/GCP/Azure, CoreWeave, Lambda Labs, OpenAI, Google DeepMind.	$6B	$27B	~35%
TOTAL AI Video Data + Training TAM	Upstream model-development spend using video data—not included in Video Compute TAM.	$15B	$50B	~27%

Key Trends in AI Video Data + Training

- Video models already account for ~20–25% of all AI training budgets, rising to 30–35% by 2030.
- Video is GPU-hour intensive, requiring far more compute than image or text training.
- Synthetic data generation (NeRFs, multi-view reconstruction, simulation) will become a major cost driver.
- Video becomes a core architecture layer for future AI systems:
    - world models rely on temporal reasoning
    - robotics requires high-frame-rate simulation
    - AI agents operate in video-like dynamic environments
    - multimodal pretraining shifts from text → images → video

These dynamics expand the total computational demand tied to video beyond traditional operational workloads.

Where Livepeer can compete in AI Video Data + Training

Although Livepeer is not designed for large-scale foundation model training, the network is well positioned to support critical upstream AI workloads such as:

- dataset preprocessing (frame extraction, slicing, deduplication, embeddings)
- batch inference and light fine-tuning
- low-cost decentralized GPU cycles
- rights-cleared video datasets with provenance
- synthetic data generation workflows
- verifiable “proof-of-compute” for data pipelines
- elastic distributed compute for episodic AI tasks

These capabilities connect Livepeer to high-growth AI markets without requiring it to become an HPC training provider.

6. Conclusion: Bridging the Video + AI Economies

Strategic Opportunity for Livepeer in Verifiable Compute & AI Data Provenance

AI-driven video pipelines increasingly require **trusted, transparent, and verifiable compute**, both during inference and during the creation of AI training datasets. A Livepeer protocol-native *Proof of Compute*, combined with a unique orchestration layer, would allow the Livepeer Network to offer **cryptographically verifiable video transformations**, covering:

- inference (captioning, detection, enhancement)
- transcoding
- generation
- dataset preprocessing
- real-time interactive compute

This verifiable layer addresses a fundamental gap in both:

- Video Compute TAM → need for verified AI-generated & AI-enhanced video
- AI Video Data + Training TAM → need for trusted preprocessing pipelines and dataset lineage for model training

No centralized cloud GPU provider currently offers decentralized, tamper-proof proofs of correctness, lineage, and compute integrity.

This positions Livepeer as the **trusted execution layer** for open video

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